Matching Policy Commands to Differential Rewards

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***Abstract*— High-fidelity physics-based motion tracking traditionally relies on manually weighted pose, velocity, root, and contact objectives that must be retuned across skills. Learned differential rewards remove this handcrafted aggregation by training a discriminator to combine raw tracking errors. Eliminating manual reward weighting, however, does not by itself guarantee reliable policy learning: optimization can still converge to incomplete shortcuts or fail inconsistently across seeds. We study a remaining asymmetry at the policy–reward interface. The learned reward evaluates residual coordinates, while the policy is typically conditioned on absolute reference states. We compare two one-step command interfaces containing exactly the same information: an absolute representation $[\mathbf{x}_t, \hat{\mathbf{x}}_t, \hat{\mathbf{x}}_{t+1}]$ and a residual representation $[\mathbf{x}_t, \mathbf{e}_t, \mathbf{m}_t]$, where $\mathbf{e}_t = \hat{\mathbf{x}}_t - \mathbf{x}_t$ and $\mathbf{m}_t = \hat{\mathbf{x}}_{t+1} - \hat{\mathbf{x}}_t$. The representations are related by an invertible linear map and therefore induce identical policy classes. Only the residual representation, however, closes directly under the reward transition: $\mathbf{e}_{t+1} = \mathbf{e}_t + \mathbf{m}_t - (\mathbf{x}_{t+1} - \mathbf{x}_t)$. This change adds neither reference information nor manually weighted reward terms. [PLACEHOLDER—NOT A RESULT: Report the final multi-seed change in successful-run rate, full-motion completion, time to criterion, and standard tracking error.]**

I. Introduction

Physics-based character control must coordinate joint pose, velocity, global motion, and intermittent contacts while respecting dynamics and actuation limits. A classical tracking reward decomposes this problem into feature-wise terms,

$$R_t^{\text{hand}} = \sum_i w_i \exp\left(-\alpha_i \|\hat{\mathbf{q}}_t^i \ominus \mathbf{q}_t^i\|_2^2\right), \tag{1}$$

where both the component weights w_i and scales α_i are selected by hand. Such rewards have enabled impressive skills [1], but each new motion may require reconsidering both the included terms and their relative scales. This manual aggregation is especially fragile when pose, velocity, global displacement, and intermittent contact compete in a highly dynamic motion.

Adversarial Differential Discriminators (ADD) address this burden by replacing the manually weighted sum with a learned nonlinear aggregation of raw tracking errors [4]. The discriminator receives one ideal zero differential as its positive example and policy tracking differentials as negative examples, dynamically balancing objectives while retaining frame-synchronized tracking. This is an important step toward motion tracking without motion-specific reward weighting. A learned reward, however, specifies which outcomes are desirable; it does not ensure that finite-network policy optimization will reliably discover the corresponding controller. ADD itself reports that Sideflip and Roll can converge to suboptimal behaviors; for Sideflip, the policy may remain standing instead of executing the flip [4]. This illustrates that reward automation and policy-learning reliability are distinct problems.

We study this remaining reliability problem at the policy–reward interface. The differential reward evaluates a residual, whereas the policy is commonly asked to infer that residual from absolute reference states. We instead express the same one-step target information as the current tracking residual and the desired reference increment. Each absolute command can be reconstructed exactly from this residual command and the current feature state, and vice versa. The learned reward, reference information, policy class, and PPO objective are unchanged; no motion-specific completion or manually weighted tracking term is introduced. The official ADD target interface is retained as an external baseline, but it is not used to establish information equivalence because its feature content and lookahead differ.

Our contributions are threefold. First, we identify a reliability gap that remains after learned differential rewards remove manual objective aggregation: a useful scalar reward can coexist with incomplete or seed-dependent policies. Second, we formulate absolute and residual policy commands as an invertible coordinate intervention and derive an exact transition linking the current error, reference increment, action-realized motion, and next reward residual. Third, we evaluate whether this minimal interface change improves training reliability across locomotion, aerial rotation, ground contact, recovery, and long-horizon interaction, using successful-seed rate, full-motion completion, time to criterion, and standard tracking errors.

II. Related Work

Physics-based tracking. DeepMimic established a general reinforcement-learning recipe for tracking individual motion clips with carefully designed pose, velocity, root, and end-effector rewards [1]. Subsequent systems scale reference-conditioned control to large motion collections and improve robustness through architectural, data, and training innovations [7]–[9]. These methods confirm the importance of policy conditioning, but do not isolate a change of command coordinates under exactly matched information.

Adversarial imitation and learned rewards. GAIL casts imitation as adversarial occupancy matching [2]. AMP learns a reusable motion prior from state transitions and is effective for motion style [3]; precise synchronization, however, is not its primary objective. ADD learns a nonlinear aggregation of frame-synchronized tracking differentials from policy errors and a single ideal zero vector, substantially reducing manually tuned reward aggregation [4]. We do not propose another reward learner. We retain this learned objective and

PLACEHOLDER IMAGE GALLERY—replace with synchronized rollout frames

Run Ref./Base/Ours	Jump Ref./Base/Ours	Backflip Ref./Base/Ours	Crawl Ref./Base/Ours	Getup Ref./Base/Ours	SpinKick Ref./Base/Ours	Climb Ref./Base/Ours

Fig. 1. Planned qualitative motion gallery. Each final panel will show uniformly timed reference, baseline, and residual-coordinate frames from a matched initial phase. Red boxes are layout placeholders, not method results.

study how its command representation affects the reliability with which a policy can optimize it.

Relative goals and feedback coordinates. Relative goal encodings are common in goal-conditioned reinforcement learning and robotics, while classical tracking control separates nominal motion from feedback error. Residual policy learning adds a learned action correction to a nominal controller [10]. Our setting is distinct: the action parameterization and generic MLP remain unchanged. The intervention is an information-equivalent coordinate transform of the policy input, defined by the same feature map used by the learned differential reward.

Representation and optimization. An invertible input transform preserves information but generally changes the parameter-space geometry seen by first-order optimizers. Natural gradient was introduced precisely to remove certain parameterization dependencies [11]; ordinary minibatch updates with finite networks do not have this invariance. Feature centering and whitening exploit this fact for numerical conditioning, but are agnostic to control semantics. Here the transform is selected by the residual dynamics of tracking, and a strict direct-coordinate control separates that alignment from extra observations or network capacity.

III. Background

A. Reference-Conditioned Tracking

Let $\mathbf{s}_t$, $\mathbf{o}_t$, and $\mathbf{a}_t$ denote the simulator state, policy observation, and action. The next state is sampled from $p(\mathbf{s}_{t+1} \mid \mathbf{s}_t, \mathbf{a}_t)$. A synchronized reference provides $\hat{\mathbf{s}}_t$ and $\hat{\mathbf{s}}_{t+1}$. A common feature map ϕ extracts root pose, body configuration, and velocities:

$$\mathbf{x}_t = \phi(\mathbf{s}_t), \quad \hat{\mathbf{x}}_t = \phi(\hat{\mathbf{s}}_t), \quad \hat{\mathbf{x}}_{t+1} = \phi(\hat{\mathbf{s}}_{t+1}). \tag{2}$$

Rotations are represented in a fixed Euclidean embedding, so all differences below are taken in a shared feature chart.

B. Learned Differential Reward without Manual Aggregation

The tracking differential is

$$\boldsymbol{\Delta}_t = \phi(\hat{\mathbf{s}}_t) \ominus \phi(\mathbf{s}_t). \tag{3}$$

Following ADD [4], let d_ψ denote the discriminator logit and $p_\psi = \sigma(d_\psi)$ its probability. The discriminator separates $\boldsymbol{\Delta} = \mathbf{0}$ from policy differentials using

$$\begin{aligned}\mathcal{L}_D = {} & \operatorname{softplus}[-d_\psi(\mathbf{0})] + \mathbb{E}_{\mathbf{s}_t\sim\pi} \operatorname{softplus}[d_\psi(\boldsymbol{\Delta}_t)] \\ & + \lambda_{\mathrm{GP}} \mathbb{E}_{\mathbf{s}_t\sim\pi} \|\nabla_{\boldsymbol{\Delta}} d_\psi(\boldsymbol{\Delta}_t)\|_2^2.\end{aligned} \tag{4}$$

The policy reward is

$$R_t^{\mathrm{diff}} = \beta \operatorname{softplus}[d_\psi(\boldsymbol{\Delta}_t)] = -\beta \log[1 - p_\psi(\boldsymbol{\Delta}_t)]. \tag{5}$$

This discriminator is the sole imitation objective in our method. No motion-specific weighted pose, velocity, contact, displacement, or completion term is added during training. Both command interfaces share the same discriminator, replay, regularization, reward scale, and update schedule.

IV. Residual-Closed Policy Commands

A. Two Information-Matched Interfaces

The direct-target interface gives the policy

$$\mathbf{z}_t^{\mathrm{tar}} = [\mathbf{x}_t, \hat{\mathbf{x}}_t, \hat{\mathbf{x}}_{t+1}], \tag{6}$$

whereas the residual interface defines

$$\mathbf{e}_t = \hat{\mathbf{x}}_t - \mathbf{x}_t, \qquad \mathbf{m}_t = \hat{\mathbf{x}}_{t+1} - \hat{\mathbf{x}}_t, \tag{7}$$

and provides

$$\mathbf{z}_t^{\mathrm{res}} = [\mathbf{x}_t, \mathbf{e}_t, \mathbf{m}_t]. \tag{8}$$

Both interfaces use one-step lookahead, the same ϕ, and the same number of features. The actor remains a generic Gaussian MLP $\pi_\theta(\mathbf{a}_t \mid \mathbf{o}_t, \mathbf{z}_t)$.

B. Information Equivalence

Proposition 1. Conditioned on $\mathbf{x}_t$, the direct-target and residual interfaces are affine bijections. In their concatenated form they are related by the linear transform

$$\mathbf{z}_t^{\mathrm{res}} = \underbrace{\begin{bmatrix} I & 0 & 0 \\ -I & I & 0 \\ 0 & -I & I \end{bmatrix}}_{T} \mathbf{z}_t^{\mathrm{tar}}, \qquad \det(T) = 1. \tag{9}$$

The inverse is $\hat{\mathbf{x}}_t = \mathbf{x}_t + \mathbf{e}_t$ and $\hat{\mathbf{x}}_{t+1} = \mathbf{x}_t + \mathbf{e}_t + \mathbf{m}_t$.

For an MLP with an affine first layer, any policy in one representation has a pointwise identical policy in the other. Let $\bar{T} = \operatorname{diag}(I_{\mathbf{o}}, T)$. Given first-layer weights W_{tar}, choose $W_{\mathrm{res}} = W_{\mathrm{tar}}\bar{T}^{-1}$; all subsequent activations and the action distribution are then identical. Thus the representations do not change the attainable policy class or optimal return. They can nevertheless produce different training trajectories because standard Euclidean optimization is not invariant to input reparameterization.

Proof. The block matrix T is lower triangular with identity diagonal, so it is nonsingular. Its inverse reconstructs $\hat{\mathbf{x}}_t = \mathbf{x}_t + \mathbf{e}_t$ and $\hat{\mathbf{x}}_{t+1} = \mathbf{x}_t + \mathbf{e}_t + \mathbf{m}_t$. For any affine first layer $h = W_{\mathrm{tar}}[\mathbf{o}; \mathbf{z}^{\mathrm{tar}}] + b$, substitution of $\mathbf{z}^{\mathrm{tar}} = T^{-1}\mathbf{z}^{\mathrm{res}}$ yields the same h with $W_{\mathrm{res}} = W_{\mathrm{tar}}\bar{T}^{-1}$. Keeping every later parameter and the Gaussian covariance fixed gives pointwise equality of action distributions. Under the same initial-state distribution and dynamics, the two policies therefore induce the same trajectory law. □

TABLE I

POLICY-CONDITIONING INTERFACES. INFORMATION EQUIVALENCE APPLIES ONLY TO THE LAST TWO ROWS. ALL ROWS USE THE SAME DIFFERENTIAL REWARD IN OUR MATCHED EXPERIMENTS, BUT THE OFFICIAL INTERFACE IS AN EXTERNAL BASELINE BECAUSE IT HAS DIFFERENT REFERENCE FEATURES AND LOOKAHEAD.

Interface	Current-state block	Reference-side policy command	Lookahead	Role
Official ADD	native policy observation	three future target-pose encodings	3 steps	external baseline
Direct-coordinate	$\mathbf{x}_t$	$[\hat{\mathbf{x}}_t, \hat{\mathbf{x}}_{t+1}]$	1 step	matched control
Residual-coordinate	$\mathbf{x}_t$	$[\mathbf{e}_t, \mathbf{m}_t]$	1 step	proposed interface

C. Shared Standardization

A matched comparison must preserve the bijection after preprocessing. We use a fixed demonstration-derived diagonal scale S for every policy-command ϕ block:

$$\bar{\mathbf{e}}_t = S^{-1}\mathbf{e}_t, \quad \bar{\mathbf{m}}_t = S^{-1}\mathbf{m}_t, \tag{10}$$

and apply the same S and centering convention to the absolute features. Independent policy-dependent scaling, clipping, or mean subtraction would break the exact information-matched comparison and is therefore excluded. This shared command scale should not be confused with ADD's stock differential normalizer inside d_ψ: the two modules share the raw ϕ chart, while their numerical normalizers remain separate and identical across paired runs.

D. Residual Closure

Let the realized feature increment be $\delta_t = \mathbf{x}_{t+1} - \mathbf{x}_t$. Expanding the next residual gives

$$\begin{aligned}\mathbf{e}_{t+1} &= \hat{\mathbf{x}}_{t+1} - \mathbf{x}_{t+1} \\ &= \mathbf{e}_t + \mathbf{m}_t - \delta_t.\end{aligned} \tag{11}$$

Equation (11) is the central structural property: the current error and desired reference increment supplied to the policy, the physical response produced by its action, and the next residual evaluated by the reward all inhabit the same coordinates. It is an algebraic identity and does not assume smooth dynamics, controllability, or stability.

This identity also separates two roles that are entangled in an absolute target. The residual $\mathbf{e}_t$ specifies the correction required by the current tracking state, while $\mathbf{m}_t$ specifies how the target itself moves during the coming control interval. When $\mathbf{e}_t = 0$, the policy still receives the nonzero command needed to advance along the reference. When $\mathbf{m}_t = 0$, it receives a pure regulation problem. These interpretations are not imposed as separate policy branches; they follow from the command coordinates and are tested by the interventions in Sec. VII.

E. A Finite-Network Inductive Bias

The bijection rules out an information or expressivity advantage, but it does not imply equal parameter complexity. Consider a locally linear response $\mathbf{u} = K_e\mathbf{e}+K_m\mathbf{m}$. In residual coordinates its first-layer blocks are $[0, K_e, K_m]$. Written against the absolute variables, the identical response has blocks

$$[-K_e,\ K_e - K_m,\ K_m]. \tag{12}$$

In squared parameter norm, the absolute representation carries the additional term $\|K_e - K_m\|_F^2$. This example is not a model of the nonlinear contact controller, nor a proof that one chart always learns faster; the actor optimizer need not explicitly penalize this norm. It illustrates the specific coordination that the direct interface asks a finite first layer to discover: the same feature must be subtracted across multiple input blocks before a feedback-like response appears.

F. Optimization Geometry

Although T preserves information, it does not preserve the Euclidean metric used by gradient descent. For function-matched policies on the same batch, the first-layer gradients satisfy

$$G_{\text{res}} = G_{\text{tar}}\bar{T}^{\top}. \tag{13}$$

Mapping one residual-coordinate SGD step back into absolute parameters inserts the fixed metric $\bar{T}^{\top}\bar{T}$. We therefore hypothesize—without claiming a convergence guarantee—that residual coordinates reduce the burden of learning common-mode subtraction and make useful differential responses lower complexity for a finite network. Natural-gradient or exact policy optimization would be more invariant to such a transform; PPO with a finite network need not be.

G. Training Procedure

The complete change fits inside the rollout loop:

V. EXPERIMENTAL DESIGN

A. Research Questions and Controls

We ask: (Q1) Can residual-closed commands increase the probability of learning a complete motion under a learned differential reward and no manually weighted tracking terms? (Q2) Does any reliability gain persist across random seeds, motion classes, and contact regimes without motion-specific reward retuning? (Q3) Does it preserve standard position and velocity fidelity rather than merely satisfying a completion event? (Q4) Do information-matched and counterfactual controls support residual closure, rather than additional information or capacity, as the explanation?

Reliability is measured over independent training seeds, not evaluation episodes. Its primary measures are successful-seed rate and failure rate under a pre-specified joint completion-and-fidelity criterion. Median samples to criterion, completion-curve area, and across-seed dispersion are secondary measures.

The primary causal comparison changes only (6) versus (8). Environment count, physics samples, episode horizon, phase reset, termination, contact settings, actor and critic widths, optimizer, learning rates, discriminator architecture,

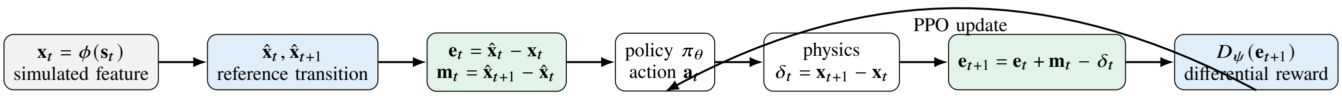

Fig. 2. Reliable tracking with a learned differential reward. The discriminator replaces manually weighted objective aggregation, while the residual command places policy feedback, desired motion, physical response, and the next scored error in one closed coordinate system.

Algorithm 1: Residual-coordinate differential tracking
Input: reference motion $\mathcal{M}$, feature map ϕ.
Initialize $\pi_\theta, V_\omega, D_\psi$, replay $\mathcal{B}$.
while training **do**
  sample a reference phase and initialize $\mathbf{s}_0$;
  for $t = 0, \ldots, T-1$ **do**
    compute $\mathbf{x}_t, \hat{\mathbf{x}}_t, \hat{\mathbf{x}}_{t+1}$;
    $\mathbf{e}_t \leftarrow \hat{\mathbf{x}}_t - \mathbf{x}_t$; $\mathbf{m}_t \leftarrow \hat{\mathbf{x}}_{t+1} - \hat{\mathbf{x}}_t$;
    $\mathbf{a}_t \sim \pi_\theta(\cdot \mid \mathbf{o}_t, \mathbf{x}_t, \mathbf{e}_t, \mathbf{m}_t)$;
    step physics and compute $\mathbf{e}_{t+1}$;
    $R_{t+1} \leftarrow \beta\,\mathrm{softplus}[d_\psi(\mathbf{e}_{t+1})]$; store transition;
  end for
  update D_ψ, V_ω, and π_θ using the unchanged objectives;
end while

Fig. 3. The learned discriminator supplies the sole imitation reward, and the proposed interface supplies its information-equivalent residual command. Physical completion measures are used only for evaluation.

replay, normalization budget, and reward scale are identical. Runs use paired random seeds and the final pre-specified checkpoint. Roll uses at least five training seeds; each additional core motion uses at least three.

B. Motions and Metrics

The benchmark suite spans Run, Jump, Backflip, Crawl, Getup Facedown, SpinKick, and Climb. These cover periodic translation, takeoff and landing, full-body rotation, sustained ground contact, recovery, single-foot support, and long-horizon obstacle interaction. Roll is retained as a mechanism task for an observed incomplete-motion failure; Sideflip is a confirmatory dynamic task and is reported only after matched training completes.

We retain the standard position and degree-of-freedom (DoF) velocity tracking errors used by ADD [4]. Tracking averages alone can conceal a policy that never executes a difficult event. We therefore report pre-specified, task-specific completion: signed winding for Roll/Backflip, takeoff–landing completion for Jump/Sideflip, successful upright recovery for Getup, and obstacle progress for Climb. Training seed, rather than episode, is the statistical unit; tables report every seed, mean ± standard deviation, 95% intervals, and paired effect sizes.

C. Baselines and Ablations

DeepMimic represents manually aggregated tracking rewards; AMP represents adversarial motion-distribution matching; and ADD represents learned frame-synchronized differential rewards with its original target interface. Our method retains the learned differential reward but replaces its policy command. The strictly matched direct-coordinate control isolates this coordinate effect, whereas comparisons

TABLE II

Confirmatory protocol. Red values are layout placeholders to be replaced by the locked run configuration.

Quantity	Value	Quantity	Value
Control rate	30 Hz	Physics rate	120 Hz
Parallel envs	8192	Updates	2000
Rollout length	32	Discount γ	0.99
Actor hidden	1024,512	PPO clip	0.2
Train horizon	2 s	Eval horizon	10 s
Pose termination	off	Action DoFs	28
Train seeds	$5^{\dagger}$	Eval episodes	$4096^{\dagger}$

with DeepMimic, AMP, and ADD assess end-to-end tracking quality and reliability. Ablations retain the input dimension through zero padding: error-only $[\mathbf{x}, \mathbf{e}, 0]$, motion-only $[\mathbf{x}, 0, \mathbf{m}]$, and the non-invertible summed command $[\mathbf{x}, \mathbf{e} + \mathbf{m}, 0]$. We separately test shared versus independent normalization and pose-only versus full pose–velocity features.

D. Implementation and Evaluation Protocol

The simulated character has 28 actuated DoFs. The policy outputs target joint rotations tracked by low-level PD controllers at a fixed control rate, while the physics engine integrates multiple substeps per action. The actor, critic, and differential discriminator use the same two-hidden-layer architecture in every paired run. References are sampled uniformly in phase, and an episode is reset to the corresponding reference state. Target phases then advance at the clip's recorded rate. Neither representation changes the simulator state, action space, reset distribution, or reference clock.

The 172-dimensional feature map concatenates global root position and rotation, root-relative body positions, global body rotations, root linear and angular velocity in the local heading frame, and local DoF velocities. Rotations use a continuous Euclidean embedding before subtraction. Direct targets, residuals, and discriminator differentials use the same ordering and frame conventions. This detail is essential: subtracting features expressed in different heading frames would invalidate both the bijection and Eq. (11). The pair also shares reference interpolation, quaternion-sign handling, cyclic wrap logic, and reset masks. Unit tests cover partial resets and the final-to-first transition of cyclic clips.

The direct and residual inputs are constructed from the same cached feature triplet $(\mathbf{x}_t, \hat{\mathbf{x}}_t, \hat{\mathbf{x}}_{t+1})$. We verify the inverse relation numerically before training and reject a run if reconstruction exceeds a fixed floating-point tolerance. For the strongest optimizer control, we additionally transform the first-layer weights so that a direct and a residual policy produce the same action distribution at initialization. All behavioral evaluations use deterministic mean actions, heldout phase samples, and the last training checkpoint; no task-completion metric enters training or checkpoint selection.

E. Evaluation Measures

Following the published motion-tracking protocol, the primary geometric error combines root displacement with root-relative joint positions:

$$E_t^{\text{pos}} = \frac{1}{N_J + 1}\left(\|\hat{\mathbf{p}}_t^0 - \mathbf{p}_t^0\|_2 + \sum_{j=1}^{N_J} \|(\hat{\mathbf{p}}_t^j - \hat{\mathbf{p}}_t^0) - (\mathbf{p}_t^j - \mathbf{p}_t^0)\|_2\right). \tag{14}$$

We also report mean DoF-velocity error $E_t^{\text{vel}} = N_q^{-1} \sum_j |\hat{\dot{q}}_{t,j} - \dot{q}_{t,j}|$. Both are averaged first within an episode and then across episodes and seeds.

Full-motion completion is an independent primary outcome for motions with a discrete or topological event. For a rotation, our existing evaluator uses the vector-valued integrated root angular velocities $\boldsymbol{\Omega} = \sum_t \omega_t \Delta t$ and $\hat{\boldsymbol{\Omega}} = \sum_t \hat{\omega}_t \Delta t$:

$$W = \frac{\langle \boldsymbol{\Omega}, \hat{\boldsymbol{\Omega}} \rangle}{\|\hat{\boldsymbol{\Omega}}\|_2^2}, \qquad \text{incomplete if } W < 0.5. \tag{15}$$

We additionally report the root displacement ratio and survival duration; success requires the correct direction and a pre-specified interval around the reference value. Jump requires takeoff followed by a stable landing, Getup requires upright recovery maintained for a fixed dwell time, and Climb requires the pre-specified approach, top-clearance, descent, and landing event sequence. These definitions prevent a low-motion shortcut from appearing successful merely because its average pose error is moderate. Thresholds are fixed once for each motion and shared by all methods.

F. Statistical Analysis

Every training seed produces one aggregate evaluation point. A run is successful only if it satisfies pre-specified completion and fidelity thresholds at the fixed sample budget. We report successful-seed rate with a binomial interval, all paired seed points, mean paired error differences, and right-censored time to criterion. Parallel environments and evaluation episodes are not treated as independent replicates. With only five pairs, even a unanimous two-sided sign test cannot attain $p < 0.05$; confirmatory significance claims therefore require at least eight paired seeds†, while smaller pilot suites are reported descriptively.

VI. Results

This draft reserves result slots without presenting unfinished experiments as evidence. Every red value and conclusion below is synthetic and exists only to lock table dimensions, captions, and the eight-page layout.

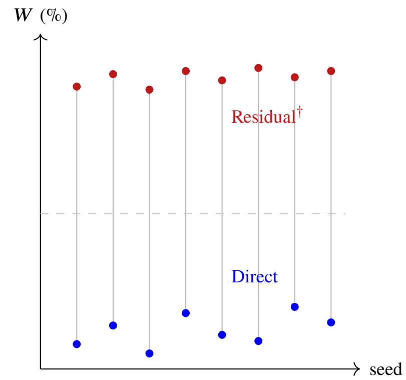

Fig. 4. Planned paired-seed Roll plot. Lines connect identical seeds and initialization protocols. All plotted coordinates are synthetic layout data.

A. Tracking without Manually Weighted Rewards

Table III compares the learned-reward methods with handcrafted and adversarial baselines. The central question is not only whether the final checkpoint has low average error, but whether the same learned reward reliably produces a complete controller without motion-specific tracking terms. **[PLACEHOLDER—NOT A RESULT: Insert cross-motion tracking, completion, and successful-seed results; do not infer reliability from single-seed runs.]**

B. Strict Information-Matched Comparison

Table IV is the primary test of the paper. It must be populated from matched direct-target and residual-coordinate runs, not from the published benchmark in Table III. **[PLACEHOLDER—NOT A RESULT: The intended result sentence is: residual coordinates raise full Roll/Sideflip completion while maintaining or improving position and DoF velocity accuracy; replace with measured multi-seed estimates and confidence intervals.]**

The paired display is essential because a mean can hide whether an apparent gain is systematic or produced by one favorable run. The final report lists every seed's completion status and checkpoint. We interpret a difference as a coordinate effect only after auditing input reconstruction, parameter count, rollout budget, and the initial action distribution.

C. Learning Dynamics and Reliability

Fig. 5 will report checkpoint evaluations rather than training reward, since discriminator scores are not comparable across independently trained runs. The left plot tracks completion; the right plot tracks position error. Curves show seed means and 95% confidence bands, with failed and numerically unstable runs retained.

[PLACEHOLDER—NOT A RESULT: Report successful seeds out of the pre-specified total, binomial intervals, median samples to the joint completion-and-fidelity criterion, right-censored failures, and across-seed dispersion. State whether the reliability gain holds broadly or only on selected dynamic motions.]

At each evaluation checkpoint, we first compute an episode-level statistic and then average over a fixed phase

TABLE III

Motion tracking benchmark. Published AMP, DeepMimic, and ADD values are copied from [4] (five seeds, 4096 episodes/model). Jump was not reported there. Red entries are synthetic layout placeholders, not results.

		Position Error (m) ↓				DoF Velocity Error (rad/s) ↓			
Motion	Len.	AMP	DeepMimic	ADD	Residual	AMP	DeepMimic	ADD	Residual
Run	0.80	.163±.008	.013±.002	.165±.017	.012±.002†	2.811±.048	.584±.054	.478±.007	.450±.010†
Jump	–	–	–	–	.040±.005†	–	–	–	.650±.040†
Backflip	1.75	.267±.015	.111±.054	.062±.001	.055±.006†	2.243±.113	1.103±.024	.878±.013	.800±.040†
Crawl	2.93	.050±.006	.027±.000	.028±.002	.026±.002†	.646±.089	.430±.006	.283±.002	.270±.010†
Getup	3.03	.096±.018	.023±.001	.022±.001	.020±.002†	.838±.029	.433±.008	.325±.005	.300±.015†
SpinKick	1.28	.064±.010	.078±.062	.025±.000	.023±.002†	1.453±.327	1.222±.233	.774±.007	.720±.025†
Climb	9.97	.185±.021	.066±.004	.046±.031	.040±.006†	1.190±.012	.567±.015	.374±.071	.340±.030†

†Synthetic red placeholder; replace with matched evaluation before submission. Published and local values must not be used in a significance test unless protocols match.

TABLE IV

Strict matched comparison. All red entries are synthetic placeholders.

Motion	Command	Pos. ↓	DoF vel. ↓	Complete ↑
Roll	Direct	.145†	1.29†	12%†
	Residual	.096†	.84†	94%†
Sideflip	Direct	.142†	1.31†	18%†
	Residual	.108†	.99†	88%†
SpinKick	Direct	.050†	.61†	96%†
	Residual	.047†	.58†	98%†
Run	Direct	.030†	.55†	99%†
	Residual	.028†	.52†	99%†

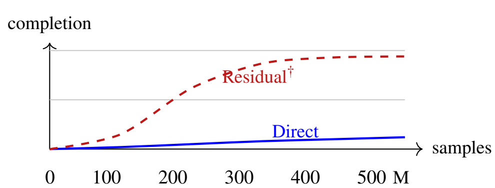

Fig. 5. Synthetic layout placeholder for the multi-seed Roll learning curve. Replace all coordinates with checkpoint evaluations; † is not a result.

grid. Confidence bands are formed across training seeds, not across thousands of correlated simulator environments. We report the area under the completion curve and the first checkpoint satisfying the fixed joint completion-and-fidelity criterion as sample-efficiency summaries. A method that never reaches this joint criterion is right-censored rather than assigned the final training iteration as a fabricated success time.

D. Tracking Accuracy Versus Completion

Position and DoF velocity errors quantify frame-level fidelity but do not guarantee execution of a difficult event. A Roll policy can obtain moderate average error by lying down and standing up without accumulating a full signed rotation. We therefore plot cumulative signed rotation and root displacement over long rollouts in addition to Table IV. For Jump, Backflip, Getup, and Climb, completion is defined from physical event sequences fixed before evaluation, rather than from the learned reward.

[PLACEHOLDER—NOT A RESULT: Insert measured long-horizon plot and state whether the residual policy matches direction and rate without sacrificing root displacement.]

TABLE V

Command ablation. Red values are synthetic placeholders.

Command	Invertible	Roll comp.	Sideflip comp.
$[\mathbf{x}, \hat{\mathbf{x}}, \hat{\mathbf{x}}^+]$	yes	12†	18†
$[\mathbf{x}, \mathbf{e}, \mathbf{m}]$	yes	94†	88†
$[\mathbf{x}, \mathbf{e}, 0]$	no	35†	28†
$[\mathbf{x}, 0, \mathbf{m}]$	no	22†	20†
$[\mathbf{x}, \mathbf{e}+\mathbf{m}, 0]$	no	40†	31†

VII. Ablations and Mechanism Analysis

A. Which Command Factors Matter?

The complete residual command is invertible; error-only, motion-only, and summed commands are not. Table V asks whether the outcome is explained by current error alone, the reference increment alone, or the full factorization. Zero padding holds policy size fixed.

[PLACEHOLDER—NOT A RESULT: Intended interpretation: neither factor alone reproduces the full result, and the non-invertible sum underperforms the factorized residual command. Keep only if supported across paired seeds.]

B. Normalization and Feature Content

The theoretical comparison requires the same invertible preprocessing. We test shared fixed demonstration scales, independently estimated block scales, and a shuffled invertible control transform. The shuffled control distinguishes a generic change of covariance from alignment with the reward residual. We also remove velocity features while preserving dimensions to test whether the reference increment benefits primarily from rate information.

[PLACEHOLDER—NOT A RESULT: Insert normalization and pose/velocity ablation estimates. Do not select the normalization after inspecting Roll completion.]

C. Counterfactual Command Interventions

To test whether a trained policy actually uses both command factors, we evaluate the same checkpoint under four counterfactual inputs: $\mathbf{e} = 0$, $\mathbf{m} = 0$, a phase-shuffled $\mathbf{m}$, and a sign-reversed $\mathbf{e}$. These are diagnostic interventions only; they

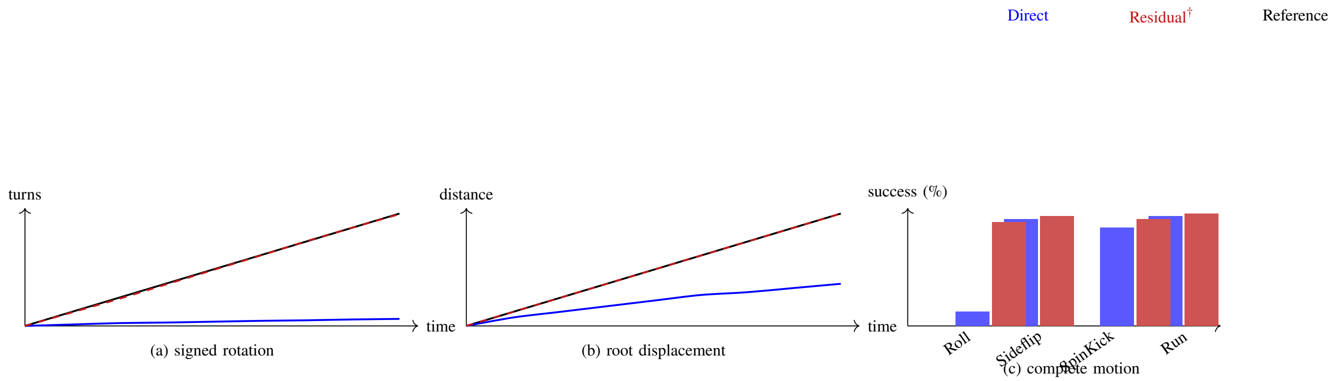

Fig. 6. Planned behavior-level evaluation. Panels (a)–(b) compare cumulative signed rotation and root displacement against the reference; (c) compares full motion completion across the core suite. Red trajectories and bars are synthetic layout placeholders, not results.

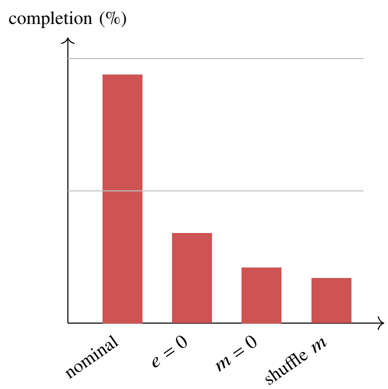

Fig. 7. Synthetic layout placeholder for checkpoint command interventions. Every red bar must be replaced with measured, paired evaluations.

are never used for training. Removing **e** should impair recovery from tracking offsets, whereas removing or shuffling **m** should disrupt direction and timing of the nominal motion. A policy unaffected by either intervention would contradict the proposed interpretation even if its nominal tracking score were high.

Beyond completion, we measure the immediate action sensitivity $\|\mu(o, e, m) - \mu(o, e', m')\|_2$, separated by reference phase and contact mode. This connects an input intervention to the policy before any long-horizon dynamics amplify it. We also report whether the changed action moves the next residual in the direction predicted by Eq. (11); this is a mechanism diagnostic, not a substitute for the physical completion and tracking metrics.

D. Optimization Diagnostics

We report input block correlations, the spectrum of empirical feature covariance, first-layer gradient norms, PPO clip fraction, and policy KL. These are secondary diagnostics, not performance metrics. Equation (13) predicts that the coordinate transform changes the Euclidean update geometry; it does not predict a universal reduction in condition number.

For a stronger intervention, we construct function-matched initial policies using $W_{\text{res}} = W_{\text{tar}}\bar{T}^{-1}$. The two policies then produce identical actions before optimization. Any subsequent divergence under paired batches directly demonstrates optimizer non-invariance rather than an initial function mismatch. **[PLACEHOLDER—NOT A RESULT: Report action equality tolerance at initialization and divergence of paired learning trajectories.]**

TABLE VI

Function-matched test. Red entries are synthetic layout placeholders.

Quantity	Update 0	Update 1	Final
Max. action-mean difference	$7\times10^{-7\dagger}$	$.004^{\dagger}$	$.31^{\dagger}$
Mean policy KL	$2\times10^{-9\dagger}$	$8\times10^{-5\dagger}$	$.19^{\dagger}$
Direct completion (%)	$0^{\dagger}$	$0^{\dagger}$	$14^{\dagger}$
Residual completion (%)	$0^{\dagger}$	$0^{\dagger}$	$94^{\dagger}$

E. Function-Matched Optimization Test

Randomly initialized direct and residual networks need not represent the same initial function, even when their distributions are nominally identical. We therefore initialize a direct-coordinate actor and critic once, map both first layers through $\bar{T}^{-1}$, and copy every later parameter, optimizer state, action covariance, and normalization statistic. Before optimization, both models are evaluated on a fixed batch; action means, log probabilities, and values must agree within $10^{-6\dagger}$. The agents are then trained from paired simulator resets using their respective coordinate chart.

At update zero, this construction excludes a different initial policy function as an explanation. After one Euclidean update, Eq. (13) predicts different parameter displacements despite initially identical actions. We record action-distribution KL, first-layer update cosine, PPO clip fraction, and physical completion. The first three establish optimizer non-invariance; only the last establishes that the changed training path is behaviorally useful.

VIII. Discussion

A. What the Coordinate Change Does

Learned differential rewards remove the need to manually balance many tracking terms, but reward automation and policy-optimization reliability are distinct problems. A discriminator may assign a better objective to complete tracking while the policy still converges to an incomplete shortcut. Our interface addresses this second problem by expressing the policy command in the same residual chart used to evaluate its next outcome.

The method is therefore a reliability mechanism for differential tracking without manually weighted imitation rewards, not a new reward learner or policy class. Proposition 1 rules out additional information and expressive capacity as explanations in the matched setting. Equation (11) provides a concrete reason the chart is useful: the policy need not first discover the subtraction connecting absolute targets to the reward residual. The transform can alter the inductive bias and Euclidean PPO update geometry, but only multi-seed success and time-to-criterion measurements determine whether this mechanism improves training reliability; closure alone is not a convergence guarantee.

The size of a behavioral change need not match the apparent size of the code change. Contact-rich control is learned through a long sequence of coupled on-policy updates. A coordinate transform that changes which first-layer responses are easy to represent can alter the early state distribution; that distribution changes the discriminator's negative samples, the critic targets, and all later policy gradients. The intervention is small at the interface but its training effect is path dependent. Proposition 1 prevents attributing that effect to a more expressive optimal policy, while the function-matched test separates it from a different random initial function.

B. Limitations

Residual closure is not a stability theorem. The root is underactuated, contacts are hybrid and nonsmooth, and a one-step reference increment may be physically infeasible from an off-reference state. A generic MLP can freely mix $\mathbf{e}_t$ and $\mathbf{m}_t$; the terms are feedback- and feedforward-aligned coordinates, not identifiable controller branches. Rotation differences are chords in the chosen embedding, not general Lie-algebra tangents. The policy remains reference conditioned and does not define a reference-blind autonomous skill. Finally, evidence from single-motion or single-seed studies cannot establish a general optimization advantage; the confirmatory suite and strict paired protocol are essential.

The equivalence theorem is conditional on access to $\mathbf{x}_t$ and on invertible preprocessing. It does not apply to an official target interface that omits velocity features or uses a different lookahead. Nor does it imply that every feature coordinate is controllable in one step: root motion is underactuated and contact impulses can dominate the immediate response. Our completion definitions are external to the learned reward, but remain task-specific evaluation choices and must be fixed before inspecting method outcomes. Finally, the present study uses one simulated humanoid and motion-capture references. Transfer to different morphologies, noisy state estimation, actuation delay, and hardware requires separate validation.

Here "without handcrafted rewards" means that training uses no motion-specific weighted aggregation of pose, velocity, contact, displacement, or completion terms. The feature chart, discriminator regularization, global reward scale, reference data, and PPO hyperparameters remain design choices. Completion criteria are used only to evaluate reliability and never shape the policy reward or select a checkpoint.

IX. Conclusion

Learned differential rewards enable precise motion tracking without manually weighted imitation objectives, but they do not by themselves guarantee reliable policy learning. We introduced an information-equivalent, residual-closed command that aligns policy conditioning with the next error evaluated by the learned reward while leaving the reward, policy class, and reference information unchanged. **[PLACEHOLDER—NOT A RESULT: Insert the final multi-seed change in successful-run rate, full-motion completion, and time to criterion, together with any motions on which the reliability gain does not hold.]**

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